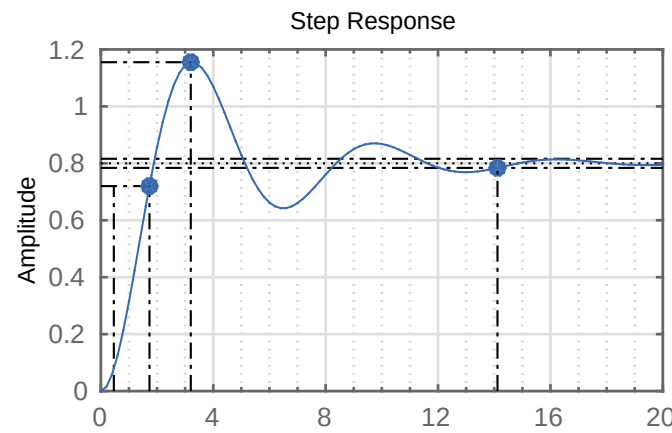


1. What is, approximately, the rise time T_r of this system?

- (a) 0.5 s
- (b) 14 s
- (c) 3 s
- (d) 1.5 s

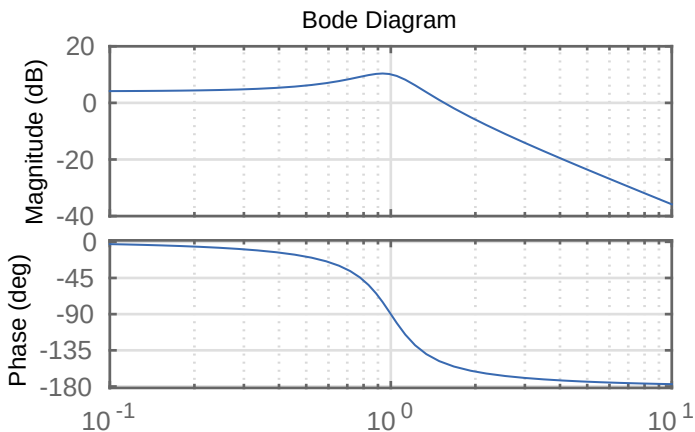


2. If $G(i\omega) \equiv 1$, what is the static gain in dB?

- (a) -3 dB
- (b) 6 dB
- (c) 0 dB
- (d) 10 dB

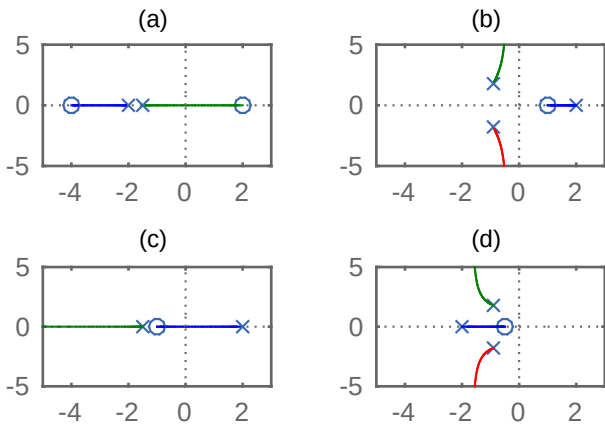
3. What is the phase margin of this system?

- (a) 30°
- (b) 150°
- (c) 60°
- (d) -60°



4. Which root locus matches the transfer function?

$$G(s) = \frac{K(s + 1)}{(s + 1.5)(s - 2) + K(s + 1)}$$



5. Which one of these is a definition of instability?

- (a) The amplitude of the output signal is much larger than that of the input signal.
- (b) The output signal grows unbounded even though the input signal does not.
- (c) The system has a large resonance peak.
- (d) The output signal is so large that the system will break down.

Answers:	
1.	
2.	
3.	
4.	
5.	